

KINETICS & REACTOR DESIGN

CHE F311 û Problems and Solutions û I-Semester 2026-27

Every numerical question in the uploaded decks, with its solution.

Every question below is quoted from one of the uploaded files, and every solution uses the method taught in the deck that set it. The questions are grouped by source deck (Parts A–I). Where a slide prints its own answer the solution reproduces it; where a slide leaves something undefined, the Notes box says what was assumed. All numbers are re-computed by `krd-answers-check.py` kept beside this file.

Question inventory

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The decks are photographs of slides; parts of some questions sit inside the pictures and are quoted as printed. Decks that repeat a question (the dimerisation of Q13 reappears in Ch5) are solved once and cross-referenced.

Part A – Mole balances and rates (Ch1-krd (1).pdf)

Q1 – Rocket engine: rates of reaction

Ch1, slide 7

Source: Ch1-krd (1).pdf, slide 7 (“Example”).

A rocket engine burns a stoichiometric mixture of hydrogen and oxygen. The combustion chamber is cylindrical, 75 cm long and 60 cm in diameter, and produces 108 kg/s of exhaust gas. If combustion is complete, find

the rate of reaction of hydrogen and of oxygen.

$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$. Chamber volume $V = \frac{\pi}{4}(0.6)^2(0.75) = 0.212 \text{ m}^3$. Exhaust = $108/18 = 6.0 \text{ kmol/s}$ of H_2O , so 6.0 kmol/s H_2 and 3.0 kmol/s O_2 react (complete combustion). With $-r_i = F_i/V$:

$$-r_{\text{H}_2} = \frac{6.0}{0.212} = 28.3 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = \frac{3.0}{0.212} = 14.1 \text{ kmol m}^{-3} \text{ s}^{-1}.$$

$$-r_{\text{H}_2} \approx 2.8 \times 10^4 \text{ and } -r_{\text{O}_2} \approx 1.4 \times 10^4 \text{ mol/(m}^3 \cdot \text{s)}.$$

Q2 – CSTR, PFR and batch for three rate laws

Ch1, slide 12

Source: Ch1-krd (1).pdf, slide 12 (“Problem”).

$A \rightarrow B$ isothermally in a continuous-flow reactor; consume 99% of A ($C_A = 0.01 C_{A0}$) with $F_{A0} = 5 \text{ mol/h}$ and $v_0 = 10 \text{ dm}^3/\text{h}$ (so $C_{A0} = 0.5 \text{ mol/dm}^3$). Rate laws: (a) $-r_A = k$, $k = 0.05 \text{ mol/(h} \cdot \text{dm}^3)$; (b) $-r_A = kC_A$, $k = 0.0001 \text{ s}^{-1} = 0.36 \text{ h}^{-1}$; (c) $-r_A = kC_A^2$, $k = 3 \text{ dm}^3/(\text{mol} \cdot \text{h})$. (d) The batch time for 99% in each case.

CSTR $V = F_{A0}X/(-r_A)_{exit}$; PFR $V = \int_0^X F_{A0} dX/(-r_A)$; batch $t = N_{A0} \int_0^X dX/(-r_A V)$ with $C_{A0} = 0.5$.

- (a) zero order: $V_{CSTR} = V_{PFR} = 5(0.99)/0.05 = 99 \text{ dm}^3$.
- (b) first order: $V_{CSTR} = \frac{v_0 X}{k(1-X)} = \frac{10(0.99)}{0.36(0.01)} = 2750 \text{ dm}^3$; $V_{PFR} = \frac{v_0}{k} \ln \frac{1}{1-X} = \frac{10}{0.36}(4.605) = 127.9 \text{ dm}^3$.
- (c) second order: $V_{CSTR} = \frac{v_0 X}{kC_{A0}(1-X)^2} = \frac{10(0.99)}{3(0.5)(10^{-4})} = 66000 \text{ dm}^3$; $V_{PFR} = \frac{v_0}{kC_{A0}} \frac{X}{1-X} = \frac{10}{1.5}(99) = 660 \text{ dm}^3$.
- (d) batch: (a) $t = C_{A0}X/k = 9.9 \text{ h}$; (b) $t = \ln(100)/0.36 = 12.8 \text{ h}$; (c) $t = \frac{X}{kC_{A0}(1-X)} = 66 \text{ h}$.

(a) 99 dm^3 both; (b) CSTR 2750 , PFR 128 dm^3 ; (c) CSTR 66000 , PFR 660 dm^3 ; (d) $9.9 / 12.8 / 66 \text{ h}$.

Part B – Conversion and reactor sizing (Ch2-krd (1).pdf)

Q3 – Sizing from the Levenspiel table

Ch2, slides 19–24

Source: Ch2-krd (1).pdf, slides 19–24.

Calculate the volumes required to achieve 80% conversion in a CSTR and a PFR when species A enters at 0.4 mol/s , given $F_{A0}/(-r_A)$ (m^3):

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$F_{A0}/(-r_A)$	0.89	1.08	1.33	2.05	3.54	5.06	8.00

CSTR: $V = \frac{F_{A0}}{-r_A} \Big|_{0.8} X = 8.00(0.8) = 6.40 \text{ m}^3$. PFR with the deck’s five-point quadrature ($h = 0.2$): $V = \frac{0.2}{3}[0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00] = 2.165 \text{ m}^3$ (deck value).

$$V_{CSTR} = 6.4 \text{ m}^3, \quad V_{PFR} = 2.165 \text{ m}^3.$$

Q4 – Two CSTRs in series

Ch2, slides 31–36

Source: Ch2-krd (1).pdf, slides 35–36.

With the same table, the first CSTR reaches $X = 0.4$; find the volume of each of two CSTRs for 80% overall.

$V_1 = 2.05(0.4) = 0.82 \text{ m}^3$; $V_2 = 8.00(0.8 - 0.4) = 3.20 \text{ m}^3$; total 4.02 m^3 — less than one CSTR (6.4 m^3), as the deck points out.

$$V_1 = 0.82 \text{ m}^3, \quad V_2 = 3.20 \text{ m}^3, \quad \text{total } 4.02 \text{ m}^3.$$

Q5 – Two PFRs in series

Ch2, slides 37–38

Source: Ch2-krd (1).pdf, slides 37–38.

Same feed; intermediate conversion 40%, final 80%.

$V_1 = \frac{0.2}{3}[0.89 + 4(1.33) + 2.05] = 0.551 \text{ m}^3$; $V_2 = \frac{0.2}{3}[2.05 + 4(3.54) + 8.00] = 1.614 \text{ m}^3$; total $2.165 \text{ m}^3 = \text{one PFR}$, as expected for PFRs in series (deck values).

$V_1 = 0.551$, $V_2 = 1.614$, total 2.165 m^3 .

Q6 – Mixed reactor, unknown stoichiometry

Ch2, slides 41–42

Source: Ch2-krd (1).pdf, slides 41–42.

A and B ($C_{A0} = 0.10$, $C_{B0} = 0.01 \text{ mol/L}$) react in a mixed reactor of $V = 1 \text{ L}$, $\tau = V/v_0 = 1 \text{ min}$. The outlet contains $C_A = 0.02$, $C_B = 0.03$, $C_C = 0.04 \text{ mol/L}$. Find the rates of reaction of A, B and C.

$r_i = (C_i - C_{i0})/\tau$: $r_A = -0.08$, $r_B = +0.02$, $r_C = +0.04 \text{ mol/(L}\cdot\text{min)}$. Ratio $-r_A : r_B : r_C = 4 : 1 : 2$, i.e. the stoichiometry consistent with the data is $4A \rightarrow B + 2C$.

$-r_A = 0.08$, $r_B = 0.02$, $r_C = 0.04 \text{ mol/(L}\cdot\text{min)}$; stoichiometry $4A \rightarrow B + 2C$.

Q7 – PFRs in two parallel branches

Ch2, slides 45–46

Source: Ch2-krd (1).pdf, slides 45–46.

Branch D: 20 L then 30 L in series; branch E: 40 L. What fraction of the feed should go to branch D?

For parallel PFR branches the efficient split keeps V/F equal: $F_D/F_E = 50/40$, so $F_D/F = 50/90 = 0.556$.

$5/9 \approx 0.56$ of the feed to branch D.

Q8 – P2-7: adiabatic $A + B \rightarrow C$, parts (a)–(e)

Ch2, slide 59

Source: Ch2-krd (1).pdf, slide 59 (“P2-7”).

$F_{A0} = 300 \text{ mol/min}$ with

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A \text{ (mol/dm}^3\cdot\text{min)}$	1.0	1.67	5.0	5.0	5.0	5.0	1.25	0.91

(a) PFR and CSTR volumes for 40%; (b) range of X where the two volumes are identical; (c) maximum X in a 105 dm^3 CSTR; (d) X for a 72 dm^3 PFR followed by a 24 dm^3 CSTR; (e) X for the reverse order.

With $g(X) = F_{A0}/(-r_A) = 300, 179.6, 60, 60, 60, 60, 240, 329.7 \text{ dm}^3$:

- (a) CSTR: $60(0.4) = 24 \text{ dm}^3$; PFR: trapezium over the table gives 71.9 dm^3 .
- (b) $X g(X) = \int_0^X g$ first holds near $X \approx 0.69$ (between 0.6 and 0.8, where the rate falls again); below it the CSTR is smaller, above it the CSTR is larger.
- (c) $X g(X) = 105 \Rightarrow X \approx 0.70$.
- (d) PFR to $72 \text{ dm}^3 \Rightarrow X_1 \approx 0.40$; then $(X_2 - 0.4)g(X_2) = 24 \Rightarrow X_2 \approx 0.64$.
- (e) CSTR: $X_1 g(X_1) = 24 \Rightarrow X_1 = 0.40$ (the high-conversion root); PFR adds $72 \text{ dm}^3 \Rightarrow X_2 \approx 0.90$.

(a) $24 \text{ dm}^3 / 71.9 \text{ dm}^3$; (b) about $X = 0.69$; (c) $X \approx 0.70$; (d) $X \approx 0.64$; (e) $X \approx 0.90$.

Part C – Rate laws and stoichiometry (Ch3-krd.pdf)

Q9 – Relative rates: NO oxidation

Ch3, slides 5–7

Source: Ch3-krd.pdf, slides 5–7 (“Example”).

$2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$; NO_2 forms at $4 \text{ mol/(m}^3\cdot\text{s)}$.

$$\frac{r_{NO}}{-2} = \frac{r_{O_2}}{-1} = \frac{r_{NO_2}}{2} \text{ gives } r_{NO} = -4 \text{ and } r_{O_2} = -2 \text{ mol}/(\text{m}^3 \cdot \text{s}), \text{ exactly as on the deck's solution slide.}$$

$$r_{NO} = -4, r_{O_2} = -2 \text{ mol}/(\text{m}^3 \cdot \text{s}).$$

Q10 – Relative rates: $2A + 3B \rightarrow 5C$

Ch3, slides 6–7

Source: Ch3-krd.pdf, slides 6–7 (“Exercise”).

$-r_A = 10 \text{ mol}/(\text{dm}^3 \cdot \text{s})$; find the rates of B and C.

$$\frac{r_A}{-2} = \frac{r_B}{-3} = \frac{r_C}{5}: r_B = \frac{-3}{-2}(-10) = -15; r_C = \frac{5}{-2}(-10) = +25 \text{ mol}/(\text{dm}^3 \cdot \text{s}) \text{ (deck values).}$$

$$r_B = -15, r_C = +25 \text{ mol}/(\text{dm}^3 \cdot \text{s}).$$

Q11 – Saponification stoichiometric table

Ch3, slides 42–44

Source: Ch3-krd.pdf, slides 42–44 (“Example”).

$3\text{NaOH} + (\text{C}_{17}\text{H}_{35}\text{COO})_3\text{C}_3\text{H}_5 \rightarrow 3\text{C}_{17}\text{H}_{35}\text{COONa} + \text{C}_3\text{H}_5(\text{OH})_3$, i.e. $3A + B \rightarrow 3C + D$, liquid phase, X = conversion of NaOH.

$\Theta_B = \frac{1}{3}$, $\Theta_C = \Theta_D = 0$ (as fed): $C_A = C_{A0}(1 - X)$; $C_B = C_{A0}(\Theta_B - \frac{X}{3})$; $C_C = C_{A0}X$; $C_D = C_{A0}\frac{X}{3}$ — the deck's table (slides 43–44), with $V = V_0$ because the mixture is liquid.

Concentrations as above; e.g. with $\Theta_B = N_{B0}/N_{A0}$ the B entry is $C_{A0}(\Theta_B - X/3)$.

Q12 – Flow-system table for $A + 2B \rightarrow C + D$

Ch3, slides 48–51

Source: Ch3-krd.pdf, slides 48–51 (“Example”).

Liquid phase, $C_{A0} = 1.8$, $C_{B0} = 6.6 \text{ kmol}/\text{m}^3$; A is the basis.

$\Theta_B = 6.6/1.8 = 3.67$. Effluent rates: $F_A = F_{A0}(1 - X)$, $F_B = F_{A0}(3.67 - 2X)$, $F_C = F_{A0}X$, $F_D = F_{A0}X$; concentrations $C_i = F_i/v_0$, e.g. $C_B = C_{A0}(3.67 - 2X)$ (deck slides 49–51).

$$C_A = C_{A0}(1 - X), C_B = C_{A0}(3.67 - 2X), C_C = C_D = C_{A0}X.$$

Q13 – Dimerisation $2A \rightarrow R$ in a MFR

Ch3, slide 61

Source: Ch3-krd.pdf, slide 61; repeated in Ch5_-krd.pdf, slide 77 (“L 5.2”).

Pure gaseous A ($C_{A0} = 100 \text{ mmol}/\text{L}$) dimerises in a MFR ($V = 0.1 \text{ L}$). Runs: $v_0 = 10.0, 3.0, 1.2, 0.5 \text{ L}/\text{h}$ with $C_A = 85.7, 66.7, 50, 33.4 \text{ mmol}/\text{L}$. Find a rate equation.

Gas phase, $\varepsilon_A = \frac{1-2}{2} = -0.5$: $C_A = C_{A0}(1 - X)/(1 - 0.5X)$ gives $X = 0.250, 0.500, 0.667, 0.800$. MFR design: $-r_A = F_{A0}X/V = v_0C_{A0}X/V$: $-r_A = 2502, 1499, 800, 400 \text{ mmol}/(\text{L} \cdot \text{h})$. Log–log slope of $(-r_A)$ vs C_A is $1.96 \approx 2$; $-r_A/C_A^2 = 0.34, 0.34, 0.32, 0.36$.

$$-r_A \approx 0.34 C_A^2 \text{ with } C_A \text{ in mmol}/\text{L} \text{ and } -r_A \text{ in mmol}/(\text{L} \cdot \text{h}) \text{ (second order in A).}$$

Q14 – Phosphine cracking PFR

Ch3, slide 63

Source: Ch3-krd.pdf, slide 63 (“Example”).

$4\text{PH}_3 \rightarrow \text{P}_4 + 6\text{H}_2$ at 649°C , first order with $k = 10 \text{ h}^{-1}$. Size a PFR at 649°C and 460 kPa for 80% conversion of $40 \text{ mol}/\text{h}$ pure PH_3 .

$$C_{A0} = P/RT = 460\,000/(8.314 \times 922.15) = 60.0 \text{ mol/m}^3; \varepsilon_A = (10 - 4)/4 = 1.5. V = \frac{F_{A0}}{kC_{A0}} \left[(1 + \varepsilon) \ln \frac{1}{1-X} - \varepsilon X \right] = \frac{40}{10 \times 60} [2.5 \ln 5 - 1.2] = 0.0667(2.419 + 0.402 - 0.402) = 0.188 \text{ m}^3.$$

$$V \approx 0.19 \text{ m}^3 \text{ (about 190 L)}.$$

Part D – Isothermal design (Ch4-_Krd.pdf)

Q15 – Ethylene-oxide batch data

Ch4, slides 17–20

Source: Ch4-_Krd.pdf, slides 17–20.

500 mL of 2 M EO mixed with 500 mL of water (0.9 wt% H₂SO₄ catalyst) at 55°C ($C_{A0} = 1.0 \text{ kmol/m}^3$ after mixing); glycol concentrations recorded vs time. Determine k ; and the batch time for 90% conversion.

$\ln(C_{A0}/C_A)$ vs t is linear; the deck's slope is $k = 0.311 \text{ min}^{-1}$ (our least-squares fit of the printed $\ln$ column gives 0.315). Then $t_{0.9} = \ln(10)/k = 7.4 \text{ min}$. The deck's scaling slide: with $k = 10^{-4} \text{ s}^{-1}$, $t_{0.9} = 2.3/10^{-4} = 23\,000 \text{ s} = 6.4 \text{ h}$.

$$k \approx 0.311 \text{ min}^{-1}; t_{90} \approx 7.4 \text{ min (and 6.4 h for } k = 10^{-4} \text{ s}^{-1}\text{)}.$$

Q16 – Ethylene glycol CSTR

Ch4, slides 29–38

Source: Ch4-_Krd.pdf, slides 29–38.

Produce 200 million lb/yr of EG at 80% conversion; 1 lbmol/ft³ EO solution mixed with an equal volume of water (0.9 wt% H₂SO₄); $k = 0.311 \text{ min}^{-1}$. (a) CSTR volume; (b) conversion of two 800-gal reactors in parallel; (c) in series.

$$F_C = 200 \times 10^6/62 \text{ lbmol/yr} = 3.226 \times 10^6/\text{yr}; F_{A0} = F_C/0.8 = 7.67 \text{ lbmol/min}; v_{A0} = 7.67 \text{ ft}^3/\text{min}, v_0 = 15.34 \text{ ft}^3/\text{min}, C_{A0} = 0.5 \text{ lbmol/ft}^3. \text{ (a) } V = \frac{F_{A0}X}{kC_{A0}(1-X)} = \frac{7.67(0.8)}{0.311(0.5)(0.2)} = 197.3 \text{ ft}^3 \text{ (deck value)}. \text{ (b) Each reactor sees } v_{A0}: \tau = \frac{800/7.48}{7.67} = 13.94 \text{ min}, Da = k\tau = 4.34, X = \frac{Da}{1+Da} = 0.81. \text{ (c) Each sees } v_0: \tau = 6.97 \text{ min}, Da = 2.17, X = 1 - \frac{1}{(1+Da)^2} = 0.90 \text{ (deck values)}.$$

$$\text{(a) } 197.3 \text{ ft}^3; \text{(b) } X \approx 0.81; \text{(c) } X \approx 0.90.$$

Q17 – Ethane cracking PFR and tube count

Ch4, slides 44–45

Source: Ch4-_Krd.pdf, slides 44–45.

300 million lb/yr ethylene from pure ethane, irreversible elementary rate law, 80% conversion, 1100 K, 6 atm; $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kcal/mol}$. How many 0.0205 ft³, 40-ft PFR tubes in parallel?

$$k(1100) = 0.072 \exp\left[\frac{82\,000}{1.987} \left(\frac{1}{1000} - \frac{1}{1100}\right)\right] = 3.07 \text{ s}^{-1}; C_{A0} = 6 \times 101\,325/(8.314 \times 1100) = 66.5 \text{ mol/m}^3; F_{A0} = 300 \times 10^6/28 \text{ lbmol/yr} = 154 \text{ mol/s}; v_0 = 2.32 \text{ m}^3/\text{s}. \text{ C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_4 + \text{H}_2: \varepsilon = 1; V = \frac{v_0}{k} \left[2 \ln \frac{1}{1-X} - X \right] = \frac{2.32}{3.07} (2.419) = 1.83 \text{ m}^3 = 64.6 \text{ ft}^3. \text{ Tubes} = 64.6/0.0205 \approx 3.2 \times 10^3.$$

$$V \approx 1.8 \text{ m}^3 (65 \text{ ft}^3); \text{ about 3150 tubes of } 0.0205 \text{ ft}^3 \text{ in parallel}.$$

The deck solves the same problem numerically in MATLAB (slides 84–95) without printing the final volume; the closed-form first-order gas-phase integral above is the method the deck derives on slides 42–43.

Q18 – PBR with pressure drop, $A + B \rightarrow 2C$

Ch4, slides 55–58

Source: Ch4-_Krd.pdf, slides 55–58.

Equimolar feed, $k = 1.5 \text{ dm}^6/(\text{mol} \cdot \text{kg} \cdot \text{min})$, $\alpha = 0.0099 \text{ kg}^{-1}$, $\varepsilon = 0$. Find X at $W = 100 \text{ kg}$.

$\frac{dX}{dW} = \frac{kC_{A0}^2}{F_{A0}}(1-X)^2(1-\alpha W)$ integrates to $\frac{X}{1-X} = \frac{kC_{A0}^2}{F_{A0}}\left(W - \frac{\alpha W^2}{2}\right)$, $p = (1-\alpha W)^{1/2}$. The deck prints $X = 0.75$ for $\alpha = 0$, which fixes the group $\frac{kC_{A0}^2}{F_{A0}} = \frac{3}{100} = 0.03 \text{ kg}^{-1}$; then $\frac{X}{1-X} = 0.03(100 - 49.5) = 1.515 \Rightarrow X = 0.60$ — the deck's two answers.

$X = 0.60$ with pressure drop, 0.75 without.

The slide does not print C_{A0} and F_{A0} separately; the combined group 0.03 kg^{-1} is implied by its own $\alpha = 0$ answer and is stated here instead of being invented.

Q19 – PBR + CSTR in series

Ch4, slides 76–77

Source: Ch4-_Krd.pdf, slides 76–77.

First order in p_A , $A \rightarrow B$ (gas), PBR with 50 kg catalyst gives 50% conversion at $P_0 = 20 \text{ atm}$; $\alpha = 0.018 \text{ kg}^{-1}$. A CSTR with the same 50 kg is added in series. (a) Which order gives the higher conversion? (b) The exit conversion; (c) pressure at the PBR exit; (d) effect of halving catalyst diameter and enlarging the pipe 50% (turbulent).

Shooting the coupled ODEs $\frac{dX}{dW} = a \frac{1-X}{1+X}p$, $\frac{dp}{dW} = -\frac{\alpha}{2p}$ to match $X(50) = 0.5$ gives $a = kP_0/F_{A0} = 0.0247 \text{ kg}^{-1}$. (c) $p = \sqrt{1-\alpha W} = \sqrt{0.1} = 0.316 \Rightarrow P = 6.3 \text{ atm}$. (b) PBR→CSTR: $X = 0.60$; CSTR→PBR: $X = 0.69$ — putting the CSTR first is better, because the PBR then runs at full pressure. (d) Turbulent $\alpha \propto D_p^{-2}D_{pipe}^{-4}$: $\alpha' = 0.018 \times 2 \times 1.5^{-4} = 0.0071$, and PBR→CSTR rises to $X \approx 0.73$ (similarly higher in the other order).

(a) CSTR upstream; (b) $X \approx 0.69$ (0.60 the other way); (c) 6.3 atm; (d) $X \approx 0.73$.

Q20 – Series reactions in a batch reactor

Ch4, slides 78–79

Source: Ch4-_Krd.pdf, slides 78–79.

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$, $k_1 = 1 \text{ h}^{-1}$, $k_2 = 2 \text{ h}^{-1}$, $C_A(0) = 5 \text{ mol/m}^3$, $C_B(0) = C_C(0) = 0$.

$C_A = 5e^{-t}$; $C_B = 5 \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t}) = 5(e^{-t} - e^{-2t})$; $C_C = 5 - C_A - C_B$. C_B peaks at $t_{max} = \frac{\ln(k_2/k_1)}{k_2 - k_1} = 0.693 \text{ h}$ with $C_B^{max} = 1.25 \text{ mol/m}^3$.

$t_{max} = \ln 2 \approx 0.69 \text{ h}$, $C_B^{max} = 1.25 \text{ mol/m}^3$; profiles as above.

Q21 – PFR pressure drop: pipe length

Ch4, slides 90–98

Source: Ch4-_Krd.pdf, slides 90–98.

$A \rightarrow B$, first order, $P_0 = 10 \text{ atm}$, $Q_0 = 0.005 \text{ m}^3/\text{s}$, 300 K, $k = 0.1 \text{ s}^{-1}$, $M = 60 \text{ g/mol}$, $\mu = 10^{-5} \text{ Pa}\cdot\text{s}$. Length for (a) 2 cm pipe, 10%; (b) 1.5 cm pipe, 10%; (c) 1.5 cm pipe, 20%?

$\rho_0 = P_0 M / RT = 24.4 \text{ kg/m}^3$, $v_0 = Q_0 / A = 15.9 \text{ m/s}$ (2 cm), $F_{A0} = 2.03 \text{ mol/s}$. Without drop, $L_{10} = \frac{v_0}{k} \ln \frac{1}{0.9} = 16.8 \text{ m}$ and $L_{20} = 35.5 \text{ m}$ (2 cm). The deck integrates the coupled dX/dz , dP/dz ODEs (slides 92–95) and reports (a) $\approx 20 \text{ m}$ for 10% in the 2 cm pipe; (b) 10% still possible in the 1.5 cm pipe ($\approx$ longer, steeper pressure fall); (c) 20% not possible — the pressure collapses before 20% is reached (slide 98: “10% conversion is possible, but 20% is not”).

(a) $\approx 20 \text{ m}$; (b) 10% achievable; (c) 20% unattainable in the 1.5 cm pipe.

Part E – Collection and analysis of rate data (Ch5_-krd.pdf)

Q22 – Trityl + methanol

Ch5, slides 12–17

Source: Ch5_-krd.pdf, slides 12–17 and 42–49.

$(\text{C}_6\text{H}_5)_3\text{CCl} + \text{CH}_3\text{OH} \rightarrow (\text{C}_6\text{H}_5)_3\text{COCH}_3 + \text{HCl}$; $C_{B0} = 0.5 \text{ M}$; trityl data ($\times 10^3 \text{ mol/dm}^3$): 50, 38, 30.6,

25.6, 22.2, 19.5, 17.4 at $t = 0, 50, \dots, 300$ min. (1) order in trityl; (2) rate constant given first order in methanol.

Finite-difference rates $-dC_A/dt$ (deck slide 17): 2.86, 1.94, 1.24, 0.84, 0.61, 0.48, 0.36×10^{-4} . Log-log slope = 1.99 ≈ 2 (deck) — second order in trityl. Integral check: $1/C_A - 1/C_{A0}$ vs t is linear with slope $k' = 0.124 \approx 0.125 \text{ dm}^3/(\text{mol}\cdot\text{min})$. With $-r_A = kC_A^2C_B$ and $C_B \approx C_{B0}$: $k = k'/0.5 = 0.25 \text{ dm}^6/(\text{mol}^2\cdot\text{min})$. The deck's nonlinear regression returns $\alpha \approx 2$ with the same k' .

Second order in trityl; $k' = 0.125 \text{ dm}^3/(\text{mol}\cdot\text{min})$, $k = 0.25 \text{ dm}^6/(\text{mol}^2\cdot\text{min})$.

Q23 – Dimethyl ether decomposition bomb

Ch5, slide 75

Source: Ch5_-krd.pdf, slide 75 (“F 5.8”).

$(\text{CH}_3)_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2 + \text{CO}$ at 504°C, constant volume; π (mmHg) = 408, 488, 562, 799 at $t = 390, 777, 1195, 3155$ s and 931 at $t = \infty$. (a) Why is π_0 missing, and estimate it; (b) order and rate constant.

(a) The bomb was already hot and reacting before the first reading; since $1 \text{ mol} \rightarrow 3 \text{ mol}$, $\pi_0 = \pi_\infty/3 = 310 \text{ mmHg}$.
(b) $p_A = (\pi_\infty - \pi)/2 = 261.5, 221.5, 184.5, 66.0 \text{ mmHg}$; the four values of $\ln(p_{A0}/p_A)/t$ are 4.4, 4.3, 4.3, $4.9 \times 10^{-4} \text{ s}^{-1}$ — first order.

$\pi_0 \approx 310 \text{ mmHg}$; first order, $k \approx 4.4 \times 10^{-4} \text{ s}^{-1}$.

Q24 – First-order reversible $A \rightleftharpoons R$

Ch5, slide 72

Source: Ch5_-krd.pdf, slide 72.

$C_{A0} = 0.5 \text{ mol/L}$, $C_{R0} = 0$; after 8 min $X = 33.3\%$; $X_e = 66.7\%$. Find the rate equation.

$\ln \frac{X_e}{X_e - X} = (k_1 + k_2)t$: $\ln 2 = (k_1 + k_2)(8) \Rightarrow k_1 + k_2 = 0.0866 \text{ min}^{-1}$; $K_e = X_e/(1 - X_e) = 2 = k_1/k_2$. Hence $k_1 = 0.0578$, $k_2 = 0.0289 \text{ min}^{-1}$ and $-r_A = 0.0578 C_A - 0.0289 C_R$ (mol/L·min).

$-r_A = 0.058 C_A - 0.029 C_R$ mol/(L·min).

Q25 – Half-order batch; $2A \rightarrow B$ volume change

Ch5, slide 73

Source: Ch5_-krd.pdf, slide 73.

(i) $A \rightarrow R$, $-r_A = 3C_A^{0.5}$ mol/(L·h), $C_{A0} = 1 \text{ mol/L}$: conversion after 1 h? (ii) Gas $2A \rightarrow B$, 80% A, constant pressure: volume falls 20% in 3 min; first-order k ?

(i) $2(\sqrt{C_{A0}} - \sqrt{C_A}) = 3t$ gives $\sqrt{C_A} < 0$ for $t > 0.667 \text{ h}$: the charge is consumed in 40 min, so $X(1 \text{ h}) = 1$. (ii) $V/V_0 = 1 + \varepsilon_A X$ with $\varepsilon_A = 0.8 \times \frac{1-2}{2} = -0.4$; $0.8 = 1 - 0.4X \Rightarrow X = 0.5$; $k = \ln \frac{1}{1-0.5}/3 = 0.231 \text{ min}^{-1}$.

(i) $X = 1$ (complete at 40 min); (ii) $k = 0.231 \text{ min}^{-1}$.

Q26 – $A \rightarrow 3R$ in a MFR (L 5.19)

Ch5, slide 74

Source: Ch5_-krd.pdf, slide 74.

Pure A at 3 atm, 30°C ($C_{A0} = 120 \text{ mmol/L}$) into a 1-L MFR; $v_0 = 0.06, 0.48, 1.5, 8.1 \text{ L/min}$ with $C_A = 30, 60, 80, 105 \text{ mmol/L}$.

$\varepsilon_A = 2$: $X = \frac{C_{A0} - C_A}{C_{A0} + 2C_A} = 0.50, 0.25, 0.143, 0.045$; $-r_A = v_0 C_{A0} X/V = 3.6, 14.4, 25.7, 44.0 \text{ mmol}/(\text{L}\cdot\text{min})$. Log-log slope = 2.00; $-r_A/C_A^2 = 0.0040$ in all four runs.

$-r_A = 0.004 C_A^2$ (C_A in mmol/L, $-r_A$ in mmol/(L·min)), i.e. $4 \text{ L}/(\text{mol}\cdot\text{min})$ in molar units.

Q27 – $A \rightarrow 2.5R$ batch, pressure data (L 3.25)

Ch5, slide 76

Source: Ch5_-krd.pdf, slide 76.

0°C, constant volume, pure A: p_A (mm) = 760, 600, 475, 380, 320, 275, 240, 215, 150 at $t = 0, 2, \dots, 14, \infty$.

$p_{A,\infty} = 150 \neq 0$: the reaction approaches equilibrium. First order towards it: $\ln \frac{p_{A0} - p_{Ae}}{p_A - p_{Ae}} = kt$ gives $k = 0.152\text{--}0.162$ at every point (fit 0.160 min^{-1}) — a first-order reversible rate equation represents the data.

First-order approach to $p_{Ae} = 150$ mm with $k \approx 0.16 \text{ min}^{-1}$.

Q28 – Reaction bomb $2A \rightarrow B$ at 100°C

Ch5, slides 78–79

Source: Ch5_-krd.pdf, slides 78–79.

Pure A at 1 atm and 25°C is jumped to 100°C; π falls from 1.14 atm at $t = 1$ min to 0.728 atm at $t = 20$ min (twelve readings, e.g. 1.04, 0.982, 0.940, 0.905, 0.870 at $t = 2.6$); after the weekend no A remains. Rate equation in mol, L, min.

$2A \rightarrow B$ with $\pi_0(100^\circ\text{C}) = 1$ atm: $p_A = 2\pi - 1$. Least squares: second order ($1/p_A$ linear) fits with $r^2 = 0.964$ vs 0.886 first order, giving $k_p = 0.074 \text{ atm}^{-1}\text{min}^{-1}$; with $p_A = C_A RT$ ($RT = 30.6 \text{ L}\cdot\text{atm}/\text{mol}$ at 373 K), $k_C = k_p RT \approx 2.3 \text{ L}/(\text{mol}\cdot\text{min})$.

Second order in A; $k \approx 0.074 \text{ atm}^{-1}\text{min}^{-1} \approx 2.3 \text{ L}/(\text{mol}\cdot\text{min})$ at 100°C.

Q29 – The gambling tables

Ch5, slide 80

Source: Ch5_-krd.pdf, slide 80.

Losses proportional to money held: table 1 halves it in 4 h, table 2 in 2 h. Starting with Rs. 1000 at both tables at once, when is Rs. 10 left?

First-order decay with $k = \ln 2/4 + \ln 2/2 = 0.520 \text{ h}^{-1}$; $t = \ln(1000/10)/k = 4.605/0.520 = 8.9 \text{ h}$.

About 8.9 hours.

Q30 – Second order: bigger MFR, equal PFR

Ch5, slide 81

Source: Ch5_-krd.pdf, slide 81.

$-r_A = kC_A^2$ gives 50% in a MFR. (i) conversion in a MFR of 6 times the volume; (ii) in a PFR of the same size.

Original: $kC_{A0}\tau = \frac{X}{(1-X)^2} = 2$. (i) $kC_{A0}\tau' = 12 = \frac{X}{(1-X)^2} \Rightarrow X = 0.75$. (ii) PFR: $kC_{A0}\tau = \frac{X}{1-X} = 2 \Rightarrow X = 2/3$.

(i) $X = 0.75$; (ii) $X = 0.67$.

Q31 – $A + 3B \rightarrow 6R$ with inerts

Ch5, slide 82

Source: Ch5_-krd.pdf, slide 82.

Feed $C_{A0} = 100$, $C_{B0} = 200$, $C_{I0} = 100$; $C_A = 40$ at the exit. Find X_A , X_B , C_B .

$X_A = 0.6$; $X_B = X_A \frac{3C_{A0}}{C_{B0}} \cdot \frac{1}{3} \times 3 = 0.9$; $C_B = 200 - 3(60) = 20 \text{ mol/L}$.

$X_A = 0.6$, $X_B = 0.9$, $C_B = 20$.

Q32 – Gaseous feed with T and π change

Ch5, slide 83

Source: Ch5_-krd.pdf, slide 83.

$T_0 = 1000$ K, $\pi_0 = 5$ atm, $C_{A0} = 100$, $C_{B0} = 200$; $A + B \rightarrow 5R$; at the exit $T = 400$ K, $\pi = 4$ atm, $C_A = 20$. Find X_A , X_B , C_B .

$$y_{A0} = 1/3, \delta = +3, \varepsilon_A = 1: C_A = C_{A0} \frac{1-X}{1+X} \frac{T_0}{T} \frac{\pi}{\pi_0} = 32 \frac{1-X}{1+X} = 20 \Rightarrow X_A = 0.231; X_B = X_A C_{A0}/C_{B0} = 0.115; C_B = 200 \frac{1-X_B}{1+X_A} \times 0.32 = 46.0.$$

$$X_A = 0.23, X_B = 0.12, C_B \approx 46.$$

Q33 – Sucrase kinetics, Michaelis–Menten

Ch5, slide 84

Source: Ch5_-krd.pdf, slide 84.

$C_{A0} = 1.0$, $C_{S0} = 0.01$ mmol/L; C_A (mmol/L) = 0.68, 0.60, 0.55, 0.38, 0.27, 0.16, 0.09, 0.04, 0.018, 0.006, 0.0025 at $t = 1..11$ h. Does Michaelis–Menten fit? Evaluate k_3 , C_M .

Integrated MM: $k_3 C_{S0} t = (C_{A0} - C_A) + C_M \ln \frac{C_{A0}}{C_A}$, linearised as $\frac{t}{\ln(C_{A0}/C_A)} = \frac{1}{k_3 C_{S0}} \frac{C_{A0} - C_A}{\ln(C_{A0}/C_A)} + \frac{C_M}{k_3 C_{S0}}$. Least squares over the eleven points: slope = 3.12 h, intercept = 1.61 h, giving $k_3 C_{S0} = 0.321 \text{ h}^{-1}$ ($k_3 \approx 32 \text{ h}^{-1}$) and $C_M \approx 0.52$ mmol/L; the points lie on the line ($r^2 > 0.99$), so the fit is reasonable.

Yes: $k_3 \approx 32 \text{ h}^{-1}$ (at $C_{S0} = 0.01$), $C_M \approx 0.5$ mmol/L.

Q34 – Reversible PFR: equilibrium then actual conversion

Ch5, slide 61

Source: Ch5_-krd.pdf, slide 61.

A PFR (2 m^3) processes aqueous feed (100 L/min) with $C_{A0} = 100$ mmol/L; $A \rightleftharpoons B$. Find the equilibrium conversion, then the actual conversion (and repeat for a CSTR).

The slide prints no rate constants; the textbook problem it reproduces (Levenspiel, the companion MFR version of which is printed with $-r_A = 0.04C_A - 0.01C_R \text{ mol/(L}\cdot\text{min)}$) supplies them: $K_e = k_1/k_2 = 4 \Rightarrow X_e = \frac{4}{5} = 0.80$. $-r_A = 0.05(C_A - 0.02)$; $\tau = V/v_0 = 20 \text{ min}$; $\ln \frac{C_{Af} - 0.02}{0.08} = -(k_1 + k_2)\tau = -1 \Rightarrow C_{Af} = 0.0494$, $X = 0.51$. (CSTR: $0.05(C_A - 0.02) \times 20 = C_{A0} - C_A \Rightarrow C_A = 0.06$, $X = 0.40$.)

$X_e = 0.80$; PFR $X \approx 0.51$; CSTR $X \approx 0.40$ (rate law quoted from the textbook original, as the slide omits it).

Q35 – $A \rightarrow R + 2S$ from total-pressure data

Ch5, slide 62

Source: Ch5_-krd.pdf, slide 62.

Pure A, constant volume; π (mmHg) = 7.5, 10.5, 12.5, 15.8, 17.9, 19.4 at $t = 0, 2.5, 5, 10, 15, 20 \text{ min}$.

$\Delta n = +2$: $p_A = \pi_0 - \frac{\pi - \pi_0}{2} = \frac{3\pi_0 - \pi}{2} = 7.5, 6.0, 5.0, 3.35, 2.3, 1.55$. $\ln(p_{A0}/p_A)/t = 0.089, 0.081, 0.081, 0.079, 0.079$ — first order, $k \approx 0.08 \text{ min}^{-1}$.

First order with $k \approx 0.08 \text{ min}^{-1}$.

Part F – Multiple reactions (Ch6_krd.pdf)

Q36 – Product distribution, three contacting patterns

Ch6, slide 32

Source: Ch6_krd.pdf, slide 32.

Aqueous $A + B \xrightarrow{k_1} R$ (desired), $A + B \xrightarrow{k_2} S$, with $r_R = 1.0 C_A^{1.5} C_B^{0.3}$, $r_S = 1.0 C_A^{0.5} C_B^{1.8} \text{ mol/(L}\cdot\text{min)}$. Equal streams of A and B at 20 mol/L each are fed ($C_{A0} = C_{B0} = 10$ after mixing); 90% conversion of A. Find C_R for (a) plug flow, (b) mixed flow, (c) plug flow with side entrances of B.

With $C_A = C_B = c$: $\varphi(R/A) = \frac{r_R}{r_R + r_S} = \frac{1}{1 + c^{0.5}}$. (a) $C_R = \int_1^{10} \frac{dc}{1 + \sqrt{c}} = 2[u - \ln(1 + u)]_1^{\sqrt{10}} = 2.86 \text{ mol/L}$. (b) $C_R = \varphi|_{c=1} (10 - 1) = 0.5 \times 9 = 4.5 \text{ mol/L}$. (c) Side entry of B holds C_B at the low exit level ($= 1$): $\varphi = \frac{c}{c+1}$,

$$C_R = \int_1^{10} \frac{c \, dc}{c+1} = 9 - \ln \frac{11}{2} = 7.30 \text{ mol/L.}$$

(a) 2.86; (b) 4.50; (c) 7.30 mol/L — mixed flow beats plug flow here, and keeping B low is best.

Q37 – Two MFRs in series

Ch6, slide 33

Source: Ch6_krd.pdf, slide 33.

$r_R = k_1 C_A^2$, $r_S = k_2 C_A$; feed $C_{A0} = 1.0$, $C_{R0} = 0$, $C_{S0} = 0.3$; $\tau_1 = 2.5$, $\tau_2 = 10$ min; first-reactor exit $C_A = 0.4$, $C_R = 0.2$, $C_S = 0.7$. Find the second-reactor exit.

From reactor 1: $k_1 = \frac{0.2/2.5}{0.4^2} = 0.5$; $k_2 = \frac{(0.7-0.3)/2.5}{0.4} = 0.4$ (the S balance checks: $0.4 \times 0.4 = 0.16 = (0.7 - 0.3)/2.5$).
Reactor 2: $0.4 - C_A = (0.5C_A^2 + 0.4C_A)10 \Rightarrow C_A = 0.0745$; $C_R = 0.2 + 0.5C_A^2(10) = 0.228$; $C_S = 0.7 + 0.4C_A(10) = 0.998$.

$C_A \approx 0.075$, $C_R \approx 0.23$, $C_S \approx 1.00$ mol/L.

Part G – Non-isothermal design (Ch7_krd.pdf)

Q38 – Adiabatic PFR: n-butane isomerisation

Ch7, slides 14–17

Source: Ch7_krd.pdf, slides 14–17.

n -butane $\rightleftharpoons$ isobutane, adiabatic liquid-phase PFR; $k = 31.1 \text{ h}^{-1}$ at 360 K, $E = 65.7 \text{ kJ/mol}$, $\Delta H_{Rx} = -6900 \text{ J/mol}$, $K_C = 3.03$ at 60°C, $C_{A0} = 9.3 \text{ kmol/m}^3$, $C_p = 141/141/161 \text{ J/(mol}\cdot\text{K)}$ (butane/isobutane/pentane). Process 163 kmol/h (90 mol% butane) to $X = 0.70$, feed at 330 K.

Adiabatic line $T = 330 + 43.3X$ (deck); $k(T) = 31.1e^{7902(1/360-1/T)}$; $K_C(T) = 3.03e^{830.3(1/333-1/T)}$; $-r_A = kC_{A0} \left[1 - \left(1 + \frac{1}{K_C}\right)X\right]$; $F_{A0} = 0.9 \times 163 = 146.7 \text{ kmol/h}$. The deck's table (e.g. $-r_A = 39.2$ at $X = 0$, $F_{A0}/-r_A = 23.29 \text{ m}^3$ at 0.7) is reproduced by these expressions; integrating $F_{A0}/(-r_A)$ to $X = 0.7$ gives $V \approx 2.0 \text{ m}^3$ (the deck's coarse Simpson table reads $\approx 2.4 \text{ m}^3$; the value is sensitive near $X_e \approx 0.72$).

$V \approx 2\text{--}2.4 \text{ m}^3$ for 70% (deck table: $F_{A0}/(-r_A) = 23.3 \text{ m}^3$ at $X = 0.7$).

Q39 – Adiabatic equilibrium with two interstage coolers

Ch7, slides 48–51

Source: Ch7_krd.pdf, slides 48–51.

$A \rightleftharpoons B$, pure A at 300 K; $K_C(298) = 10^5$, $\Delta H_{Rx} = -20\,000 \text{ cal/mol}$ (as implied by the deck's $K_C(T)$ and adiabatic line), $C_{pA} = C_{pB} = 50 \text{ cal/(mol}\cdot\text{K)}$, so $X_{ad} = 2.5 \times 10^{-3}(T - 300)$. Two coolers return the stream to 350 K; 95% of the equilibrium conversion in each reactor; $F_{A0} = 40 \text{ mol/s}$; cooling water at 270 K, $c_{pC} = 18 \text{ cal/(mol}\cdot\text{K)}$, $U = 100 \text{ cal/(s}\cdot\text{m}^2\cdot\text{K)}$.

$K_C(T) = 10^5 \exp[-33.78(T - 298)/T]$, $X_e = K_C/(1 + K_C)$. Reactor 1: $2.5 \times 10^{-3}(T - 300) = 0.95X_e(T) \Rightarrow T_1 = 459 \text{ K}$, $X_1 = 0.40$. Cool to 350 K. Reactor 2: $X_1 + 2.5 \times 10^{-3}(T - 350) = 0.95X_e(T) \Rightarrow T_2 = 440 \text{ K}$, $X_2 = 0.62$. Duties: $\dot{Q} = \dot{n}C_p(T_{out} - 350)$: $Q_1 = 40 \times 50 \times 109 = 218 \text{ kcal/s}$; $Q_2 = 40 \times 50 \times 90 = 179 \text{ kcal/s}$ (water flow = $\dot{Q}/(18\Delta T)$ as sized by the deck).

$X \approx 0.40$ after reactor 1 and 0.62 after reactor 2; $\dot{Q}_1 \approx 218$, $\dot{Q}_2 \approx 179 \text{ kcal/s}$.

Q40 – Propylene-oxide CSTR, adiabatic

Ch7, slides 57–59

Source: Ch7_krd.pdf, slides 57–59 (and 69–72).

300-gal glass-lined CSTR; $F_{A0} = 43.04 \text{ lbmol/h PO}$; streams: $46.62 \text{ ft}^3/\text{h PO}$, $46.62 \text{ ft}^3/\text{h methanol}$, $233.1 \text{ ft}^3/\text{h water}$ (0.1 wt% H_2SO_4); methanol 71.87 and water 802.8 lbmol/h; $T_0 = 75^\circ\text{F}$ = 535 R after the 17°F heat-of-mixing rise; first order in PO with $k = 16.96 \times 10^{12} e^{-32\,400/RT} \text{ h}^{-1}$; limit 125°F.

$v_0 = 326.3 \text{ ft}^3/\text{h}$, $C_{A0} = 0.132 \text{ lbmol}/\text{ft}^3$, $\tau = (300/7.48)/326.3 = 0.123 \text{ h}$. Energy balance: $\sum \Theta_i C_{p,i} \approx 404 \text{ Btu}/(\text{lbmol} \cdot \text{R})$, $-\Delta H_{Rx} = 36\,400 \text{ Btu}/\text{lbmol} \Rightarrow T = 535 + 90.1X$. Solving with $X = \frac{k\tau}{1+k\tau}$: $X = 0.84$, $T = 611 \text{ R} = 151 \text{ }^\circ\text{F}$ (deck table intersection $\approx 613 \text{ R}$, $X \approx 0.85$). $151 > 125 \text{ }^\circ\text{F}$: the reactor *cannot* be used adiabatically.

$X \approx 0.84\text{--}0.85$ at $T \approx 611\text{--}613 \text{ R} (\approx 151 \text{ }^\circ\text{F}) > 125 \text{ }^\circ\text{F}$ — not acceptable.

Q41 – The same CSTR with a cooling coil

Ch7, slides 60–74

Source: Ch7-_krd.pdf, slides 60–74.

A 40 ft^2 coil, $U = 100 \text{ Btu}/(\text{h}\cdot\text{ft}^2\cdot^\circ\text{F})$, coolant at $85 \text{ }^\circ\text{F}$ (545 R). Does the $125 \text{ }^\circ\text{F}$ constraint now hold?

Energy balance adds $UA(T - T_a) = 4000(T - 545) \text{ Btu}/\text{h}$: $X = \frac{17\,379(T-535)+4000(T-545)}{43.04 \times 36\,400}$ together with the mole balance. Intersection: $X \approx 0.35\text{--}0.36$, $T \approx 562\text{--}564 \text{ R} = 102\text{--}104 \text{ }^\circ\text{F}$ — the deck's fsolve gives $X = 0.364$, $T = 563.7 \text{ R}$. The constraint is satisfied with the coil.

Yes: $X \approx 0.36$ at $T \approx 564 \text{ R} (\approx 104 \text{ }^\circ\text{F}) < 125 \text{ }^\circ\text{F}$.

Q42 – Parallel reactions in a PFR with heat effects

Ch7, slide 95

Source: Ch7-_krd.pdf, slide 95.

$A \rightarrow B$ ($r_B = k_1 C_A$), $2A \rightarrow C$ ($r_C = k_2 C_A^2$); pure A at $100 \text{ mol}/\text{s}$, $150 \text{ }^\circ\text{C}$, $C_{A0} = 0.1 \text{ mol}/\text{dm}^3$; $\Delta H_1 = -20\,000$, $\Delta H_2 = -60\,000 \text{ J}/\text{mol A}$; $C_{pA} = C_{pB} = 90$, $C_{pC} = 180 \text{ J}/(\text{mol}\cdot^\circ\text{C})$; $E_1/R = 4000 \text{ K}$, $E_2/R = 9000 \text{ K}$; $k_1 = 10$, $k_2 = 0.09$ (as printed, per min). Determine the temperature and flow-rate profiles down the reactor.

Coupled ODEs (per the deck's PFR heat-effects balances): $\frac{dF_A}{dV} = -(r_1 + 2r_2)$, $\frac{dF_C}{dV} = r_2$, $\frac{dT}{dV} = \frac{r_1(-\Delta H_1) + r_2(-\Delta H_2)}{F_A C_{pA} + F_B C_{pB} + F_C C_{pC}}$ with $C_A = F_A/\dot{V}_0$, $\dot{V}_0 = 60\,000 \text{ dm}^3/\text{min}$ (taking the printed rate constants as per minute). Integrating to $V = 10 \text{ m}^3$: $F_A \rightarrow 0$, $F_C \rightarrow 0$ (B dominates), and T rises to $\approx 645 \text{ K}$ — the adiabatic limit for complete conversion to B ($\Delta T = 100 \times 20\,000/(100 \times 90) \approx 222 \text{ K}$ above 423 K).

Both reactions complete within the first m^3 ; $T \rightarrow \approx 645 \text{ K}$, product essentially B.

The slide states the problem and the data but no units for k_1, k_2 beyond “min”; the profile above takes them as printed (per minute). The qualitative result (fast exothermic run to the adiabatic temperature rise) is unaffected.

Part H – Catalysis (Catalysis_Krd.pdf)

Q43 – Pellet concentration profile

Catalysis, slide 48

Source: Catalysis_Krd.pdf, slide 48 (“Problem”).

$A \rightarrow B$, first order, spherical particle $d_p = 2 \times 10^{-3} \text{ cm}$; $C_{As} = 0.001 \text{ gmol}/\text{m}^3$; $D_e = 0.1 \text{ cm}^2/\text{s}$. The concentration halfway to the centre is $1/10$ of C_{As} . (a) C_A at $3 \times 10^{-4} \text{ cm}$ from the external surface; (b) diameter for $\eta = 0.8$.

Sphere, first order: $\frac{C_A}{C_{As}} = \frac{R \sinh(\phi r/R)}{r \sinh \phi}$. At $r = R/2$: $0.1 = \frac{2 \sinh(\phi/2)}{\sinh \phi} = \frac{1}{\cosh(\phi/2)} \Rightarrow \phi = 2 \text{ arcosh}(10) = 5.99$.
(a) $r = R - 3 \times 10^{-4} = 7 \times 10^{-4} \text{ cm}$: ratio $= \frac{1}{0.7} \frac{\sinh(0.7 \times 5.99)}{\sinh 5.99} = 0.237$, $C_A = 2.4 \times 10^{-4} \text{ gmol}/\text{m}^3$. (b) $\eta = \frac{3}{\phi^2}(\phi \coth \phi - 1) = 0.8 \Rightarrow \phi_{0.8} = 2.04$; $\phi \propto d_p$: $d_{new} = 2 \times 10^{-3} \times \frac{2.04}{5.99} = 6.8 \times 10^{-4} \text{ cm}$.

(a) $C_A \approx 2.4 \times 10^{-4} \text{ gmol}/\text{m}^3$; (b) $d_p \approx 6.8 \times 10^{-4} \text{ cm}$.

Q44 – Internal diffusion: 100 vs 150 °C

Catalysis, slide 57

Source: Catalysis_Krd.pdf, slide 57.

First order, $R = 0.2 \text{ cm}$, $D_e = 0.015 \text{ cm}^2/\text{s}$; $k(100 \text{ }^\circ\text{C}) = 0.93 \text{ s}^{-1}$, $E = 20 \text{ kcal}/\text{mol}$; $C_{Ab} = 3.25 \times 10^{-5} \text{ mol}/\text{L}$. (a) rate at $100 \text{ }^\circ\text{C}$; (b) at $150 \text{ }^\circ\text{C}$; (c) apparent activation energy.

(a) $\phi_1 = R\sqrt{k_1/D_e} = 1.575$; $\eta_1 = 0.866$; $-r_A = \eta_1 k_1 C_{Ab} = 2.6 \times 10^{-5}$ mol/(L·s). (b) $k_2 = 0.93e^{\frac{20,000}{1.987}(1/373 - 1/423)} = 22.5$ s $^{-1}$; $\phi_2 = 7.75$; $\eta_2 = 0.337$; $-r_A = 2.5 \times 10^{-4}$ mol/(L·s). (c) From the two rates, $E_{app} = R \ln(r_2/r_1)/(1/T_1 - 1/T_2) \approx 14.1$ kcal/mol — between E and $E/2$, moving toward the diffusion-limited $E/2 = 10$ kcal/mol as ϕ grows.

(a) 2.6×10^{-5} ; (b) 2.5×10^{-4} mol/(L·s); (c) $E_{app} \approx 14$ kcal/mol.

Part I – Residence time distributions (Non_ideal_KRD.pdf)

Q45 – Tracer pulse: $E(t)$ and the 3–6 min fraction

Non-ideal, slides 17–19

Source: Non_ideal_KRD.pdf, slides 17–19 (“Problem”).

C (g/m 3) = 0, 1, 5, 8, 10, 8, 6, 4, 3, 2.2, 1.5, 0.6, 0 at $t = 0, 1, \dots, 10, 12, 14$ min. Construct $E(t)$ and find the fraction that spent 3–6 min in the reactor.

$\int_0^\infty C dt = 47.4 + 2.6 = 50.0$ g·min/m 3 (Simpson on 0–10 plus 10–14, as on the deck). $E(t) = C(t)/50$: peak $E(4) = 0.20$. Fraction between 3 and 6 min: $\int_3^6 E dt \approx \frac{1}{50} \times \frac{3}{8}(8 + 3 \cdot 10 + 3 \cdot 8 + 6) = 0.465 \approx 0.5$. Mean residence time $t_m = \int tE dt \approx 5.2$ min.

$\int C dt = 50$ g·min/m 3 ; about half the material (0.47–0.51) spent 3–6 min inside; $t_m \approx 5.2$ min.

Q46 – Dispersion model: conversion four ways

Non-ideal, slides 52–59

Source: Non_ideal_KRD.pdf, slides 52–59 (“Problem”).

First order $A \rightarrow B$ in a 10 cm, 6.36 m tubular reactor, $k = 0.25$ min $^{-1}$; tracer test of Q45. Conversion by (i) closed-vessel dispersion model, (ii) PFR, (iii) single CSTR, (iv) tanks in series.

$t_m = 5.13$ min, $\sigma^2 = 5.92$ min 2 (deck values), $\sigma_\theta^2 = 0.225$; $\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2}(1 - e^{-Pe}) \Rightarrow Pe = 7.7$. $Da = kt_m = 1.28$, $q = \sqrt{1 + 4Da/Pe} = 1.29$: $X = 1 - \frac{4qe^{Pe/2}}{(1+q)^2 e^{Pe q/2} - (1-q)^2 e^{-Pe q/2}} = 0.68$ (deck value). (ii) $X = 1 - e^{-Da} = 0.72$; (iii) $X = \frac{Da}{1+Da} = 0.56$; (iv) $n = t_m^2/\sigma^2 = 4.45$: $X = 1 - (1 + Da/n)^{-n} = 0.68$.

(i) 0.68; (ii) 0.72; (iii) 0.56; (iv) 0.68.

Every number above is recomputed by `krd-answers-check.py` (kept in this folder) from the data quoted here; discrepancies with a deck’s own printed answer, where any, are discussed in that question’s Notes/Warning box.